MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE DBG

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.4 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DBG.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/158-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I(see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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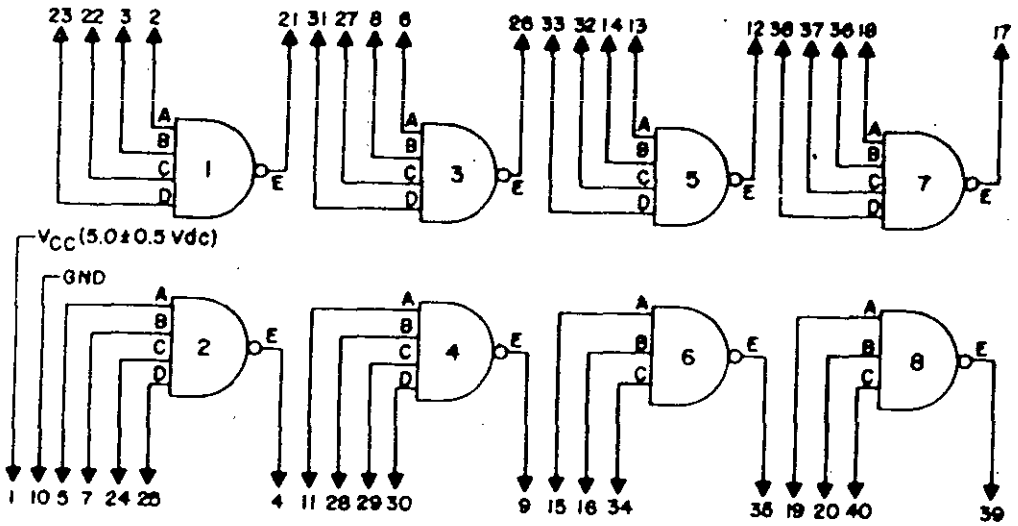


FIGURE 1. DBG module function diagram, NAND gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptable requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	V_{OUT} (low-loaded)	III	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
B1-B30	Input breakdown current	IV	S	0	100	0	100	0	100	0	100	µA
C	Isolation, frame	V	S	10	-	1	-	10	-	1	-	MΩ
D1-D8	Output voltage, low level	VI	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
E1-E30	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F30	Input current, low level	VIII	X	50	200	50	200	50	200	50	200	µA
G1-G30	Input current, high level	IX	X	0	10	0	10	0	10	0	10	µA
H1-H8	Output short circuit current	X	X	3	15	3	15	3	15	3	15	mA
J1	Power supply current, low level	XI	S	0.5	4.1	0.5	4.1	0.5	4.1	0.5	4.1	mA
J2	Power supply current, high level	XI	S	0.2	1.6	0.2	1.6	0.2	1.6	0.2	1.6	mA
K1-K8	Transition time (t_{TLH}), C_L = 40 pF	XII	S	5	60	5	60	5	60	5	60	ns
K9-K16	Transition time (t_{TLH}), C_L = 140 pF	XII	Q	10	100	10	100	10	100	10	100	ns
L1-L8	Transition time (t_{THL}), C_L = 40 pF	XIII	S	2	30	2	30	2	30	2	30	ns
L9-L16	Transition time (t_{THL}), C_L = 140 pF	XIII	Q	5	60	5	60	5	60	5	60	ns
M1-M8	Propagation delay time (t_{PLH}), C_L = 40 pF	XIV	S	10	60	10	60	10	70	10	70	ns
M9-M16	Propagation delay time (t_{PLH}), C_L = 140 pF	XIV	Q	10	90	10	90	10	100	10	100	ns
N1-N8	Propagation delay time (t_{PHL}), C_L = 40 pF	XV	S	10	66	10	66	10	70	10	70	ns
N9-N16	Propagation delay time (t_{PHL}), C_L = 140 pF	XV	Q	10	96	10	96	10	100	10	100	ns
P1-P8	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
R1-R8	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVIII	Q	0	300	0	300	0	300	0	300	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptable requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T	Power supply filter capacitance	XIX	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
U1-U30	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF

NOTES:

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through T shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/158-1 supersedes part number M28787/158. The two parts are interchangeable.

6.4 Module description. The following electrical parameter reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	
All outputs at low level	23 mW (outputs not loaded)
All outputs at high level	9 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum

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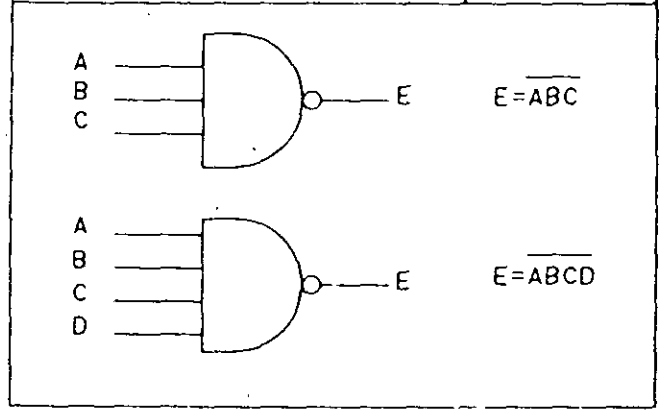
Input current	
Low level (V_{IL} = 0.0 V)	-0.2 mA maximum
(V_{IL} = 0.3 V)	-0.18 mA maximum
High level (V_{IH} = 2.4 V)	10 µA maximum
(V_{IH} = 5.5 V)	100 µA maximum
Output voltage	
Low level (I_{OL} = 2.0 mA)	0.3 V maximum
High level (I_{OH} = -100 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.3 V)	2.0 mA
High level (V_{OH} = 2.4 V)	-100 µA

6.5 Module function. This module contains two independent low power 3-input and six independent low power 4-input NAND gates.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

INPUT	OUTPUT
At least one input in low state	High state
All inputs in high state	Low state



6.5.2 Parallel outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

The circuit groups which can be paralleled are:

(1) Circuits 1 and 2.
(2) Circuits 3 and 4.
(3) Circuits 5 and 7.
(4) Circuits 6 and 8.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

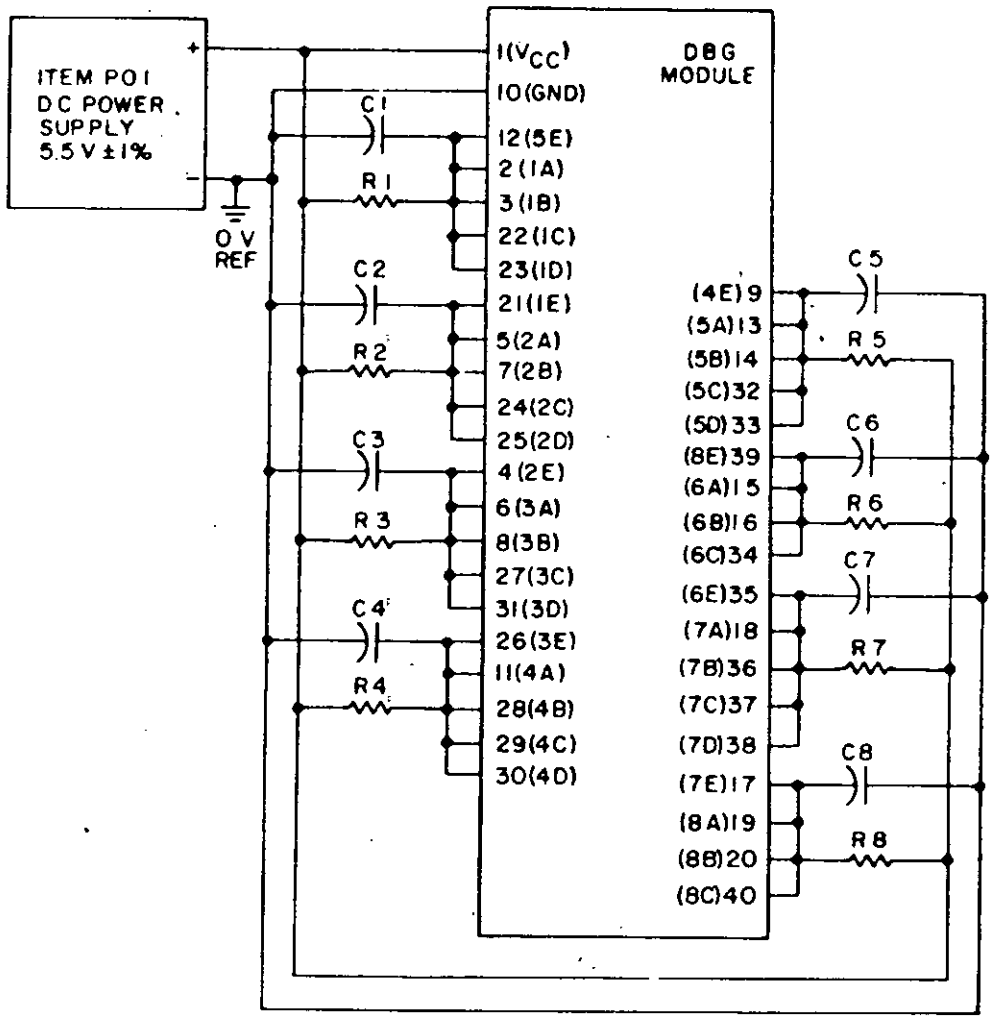
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NOTES:
1. R1-R8 = 2.75 kΩ ±1%, 1/8 W.
2. C1-C8 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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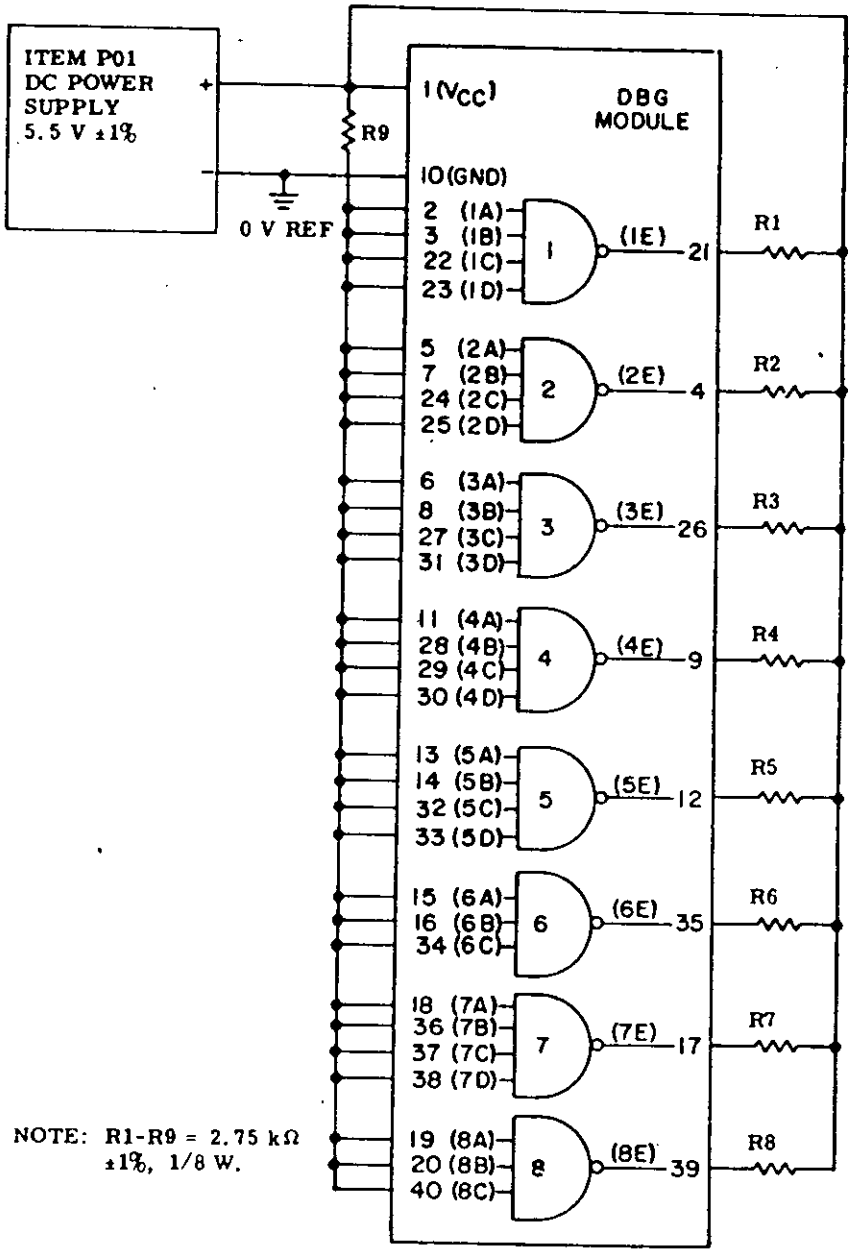


FIGURE 3. Component temperature test circuit.

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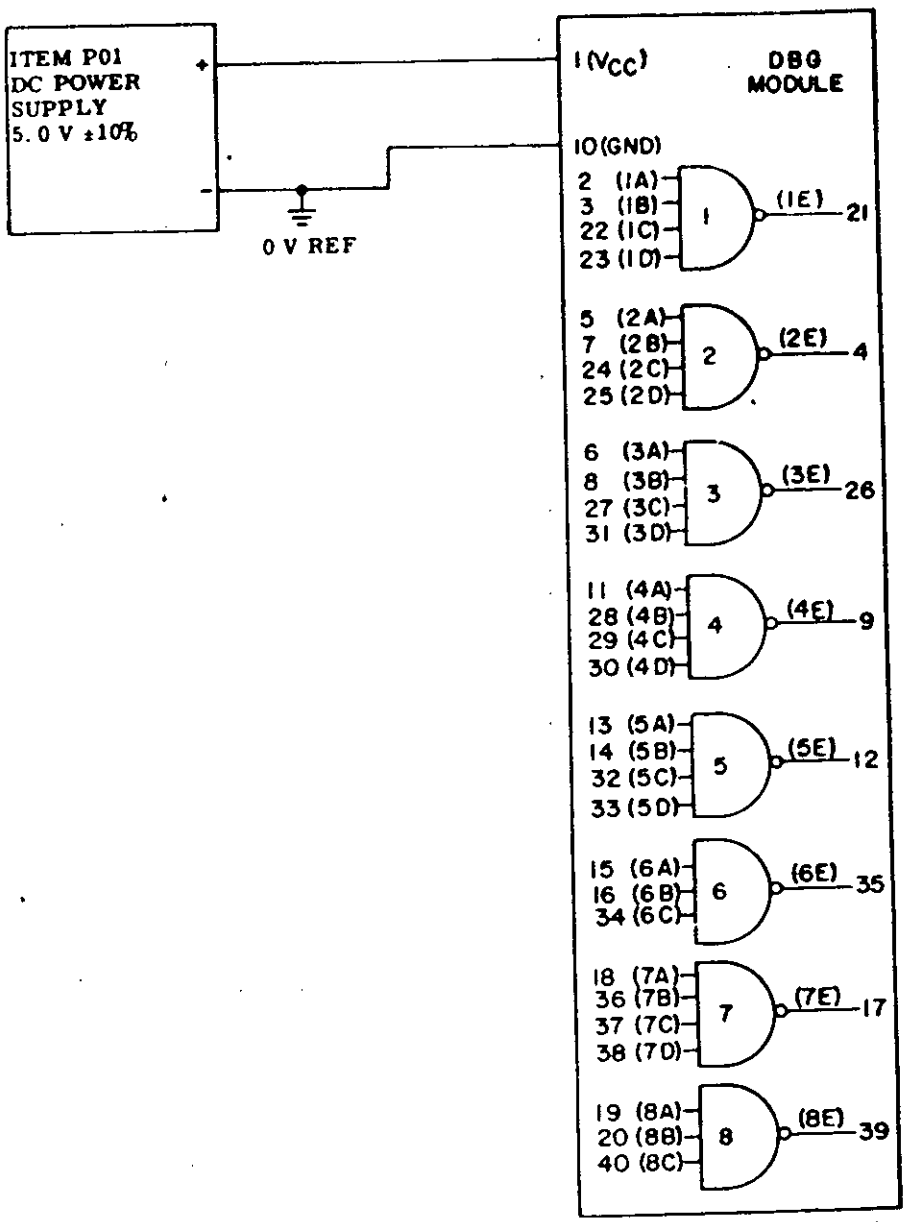


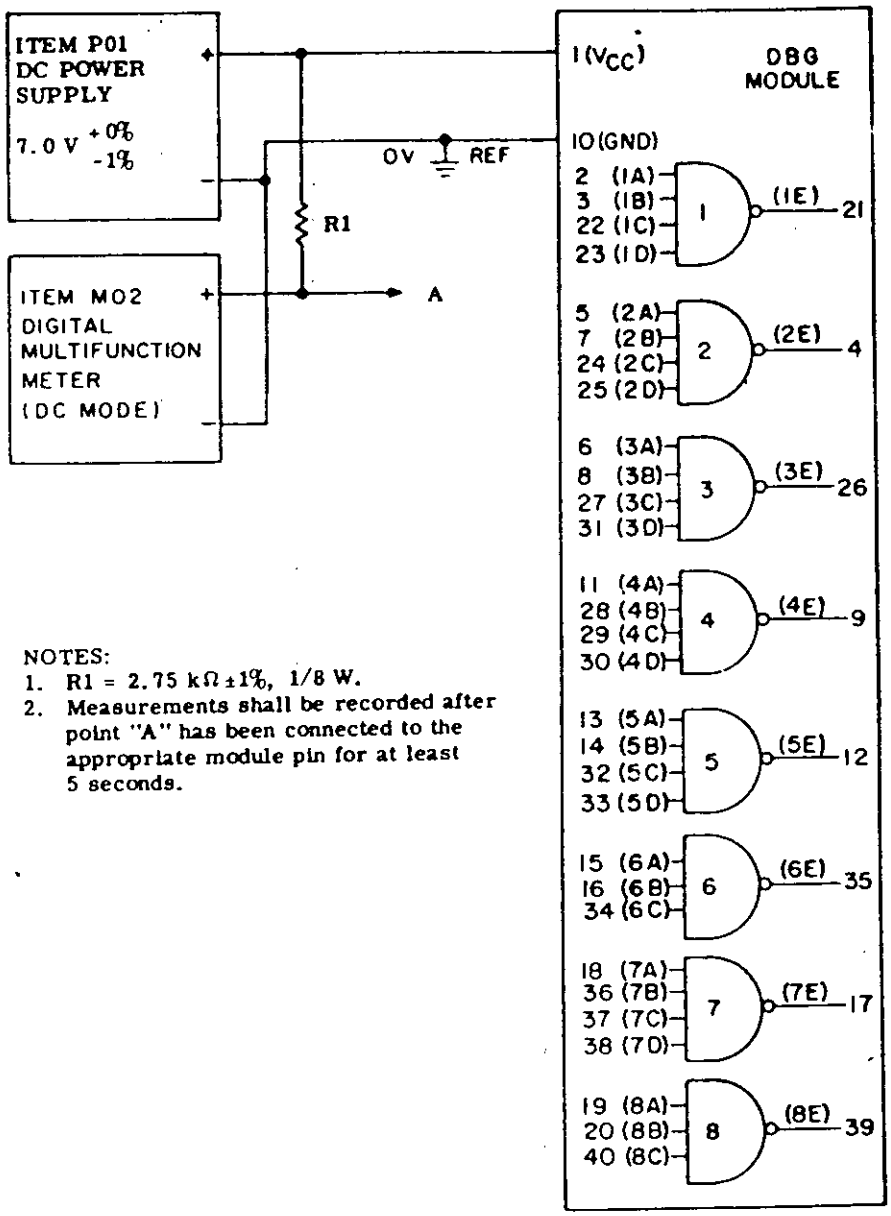
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1E V_{OUT} (low loaded)	21	0.0	0.6	V	5
A2	2E	4	↓	↓	↓	↓
A3	3E	26				
A4	4E	9				
A5	5E	12				
A6	6E	35				
A7	7E	17				
A8	8E	39				

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NOTES:
1. R1 = 2.75 kΩ ±1%, 1/8 W.
2. Measurements shall be recorded after point "A" has been connected to the appropriate module pin for at least 5 seconds.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input breakdown current	2	3, 22, 23	0	100	μA	6
B2	1B	3	2, 22, 23				
B3	1C	22	2, 3, 23				
B4	1D	23	2, 3, 22				
B5	2A	5	7, 24, 25				
B6	2B	7	5, 24, 25				
B7	2C	24	5, 7, 25				
B8	2D	25	5, 7, 24				
B9	3A	6	8, 27, 31				
B10	3B	8	6, 27, 31				
B11	3C	27	6, 8, 31				
B12	3D	31	6, 8, 27				
B13	4A	11	28, 29, 30				
B14	4B	28	11, 29, 30				
B15	4C	29	11, 28, 30				
B16	4D	30	11, 28, 29				
B17	5A	13	14, 32, 33				
B18	5B	14	13, 32, 33				
B19	5C	32	13, 14, 33				
B20	5D	33	13, 14, 32				
B21	6A	15	16, 34				
B22	6B	16	15, 34				
B23	6C	34	15, 16				
B24	7A	18	36, 37, 38				
B25	7B	36	18, 37, 38				

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B26	7C Input breakdown current	37	18, 36, 38	0	100	µA	6
B27	7D	38	18, 36, 37	↓	↓	↓	↓
B28	8A	19	20, 40				
B29	8B	20	19, 40				
B30	8C ↓	40	19, 20				

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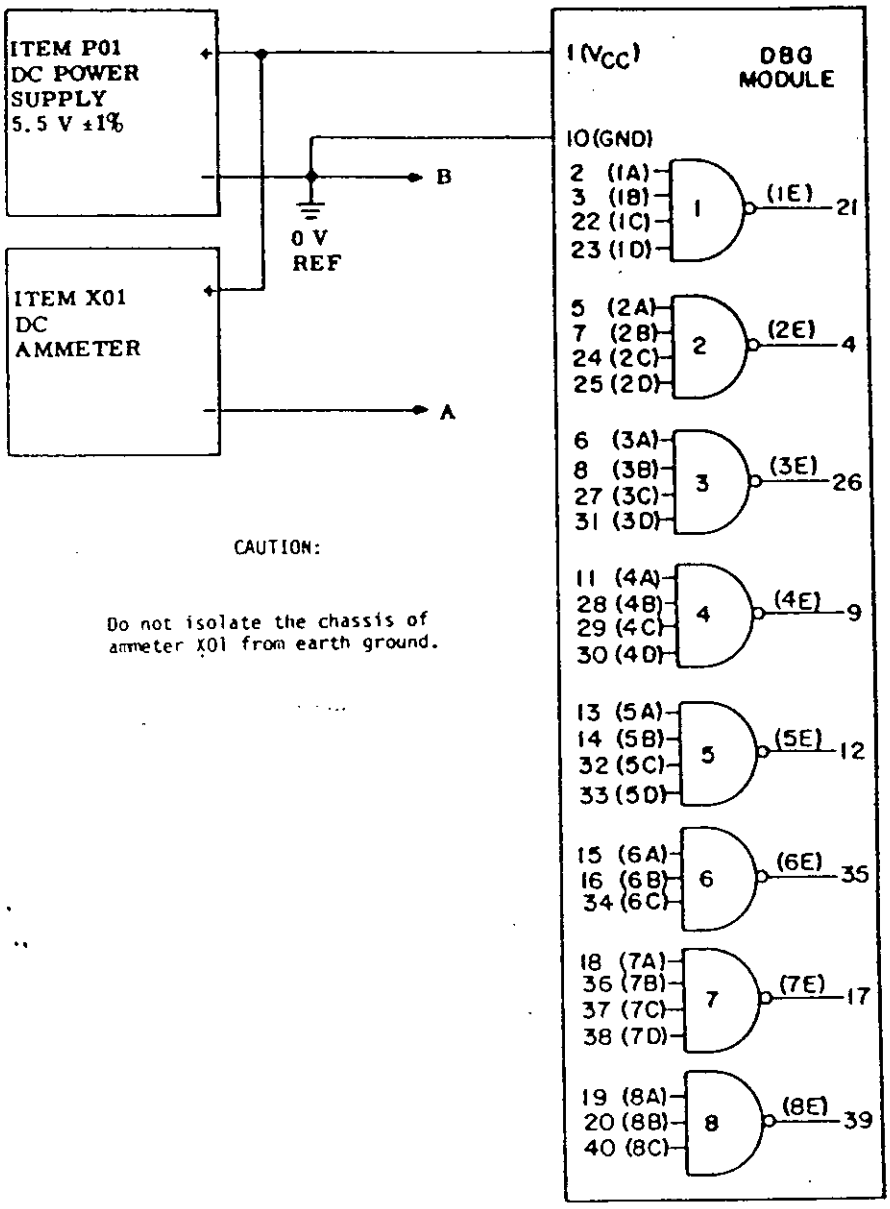


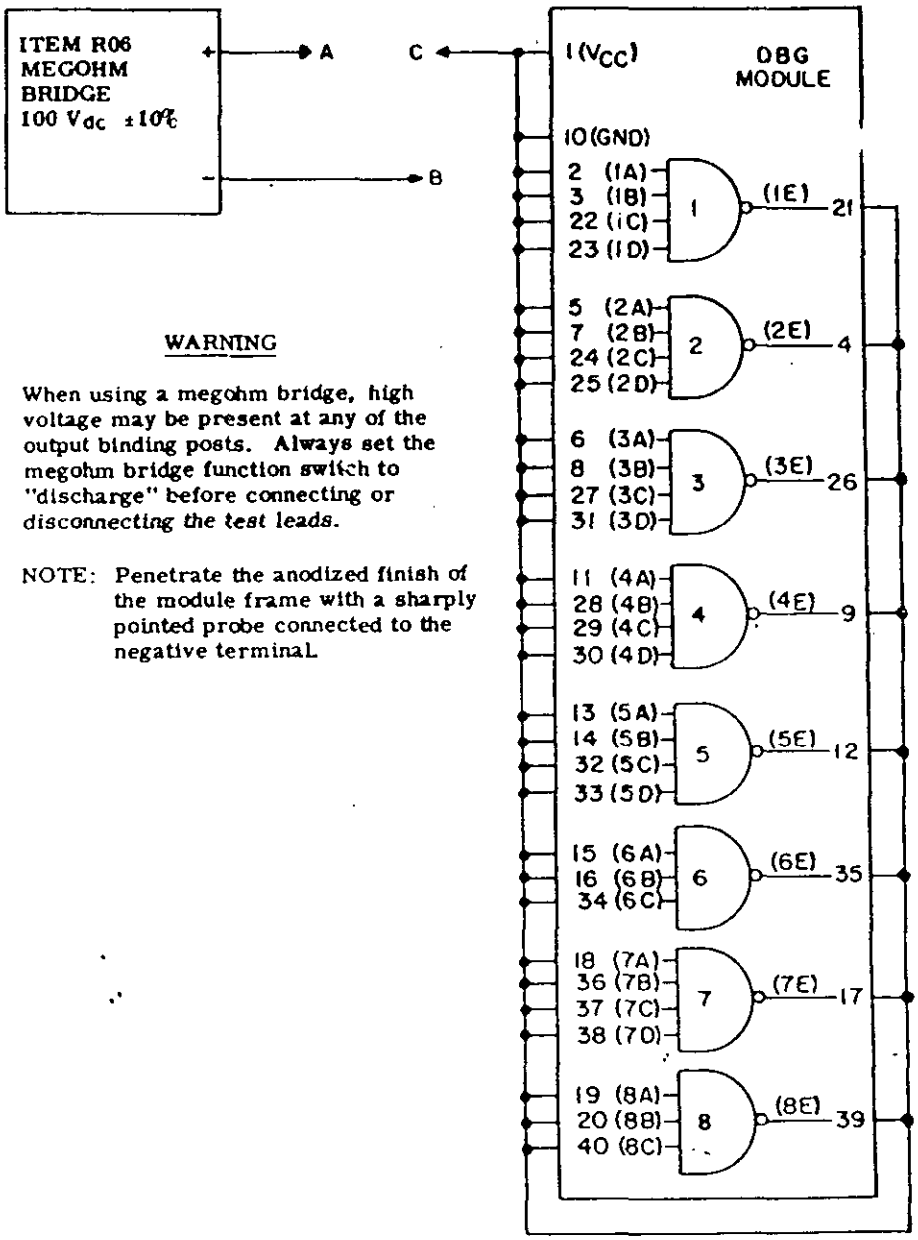
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation, frame	C	Frame	10	-	MΩ	7

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1E Output voltage, low level	2,3,22,23	Open	21	0.0	0.3	V	8
D2	2E	5,7,24,25		4				
D3	3E	6,8,27,31		26				
D4	4E	11,28,29,30		9				
D5	5E	13,14,32,33		12				
D6	6E	15,16,34		35				
D7	7E	18,36,37,38		17				
D8	8E	19,20,40		39				

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TABLE VII. Output voltage (high-level) test.

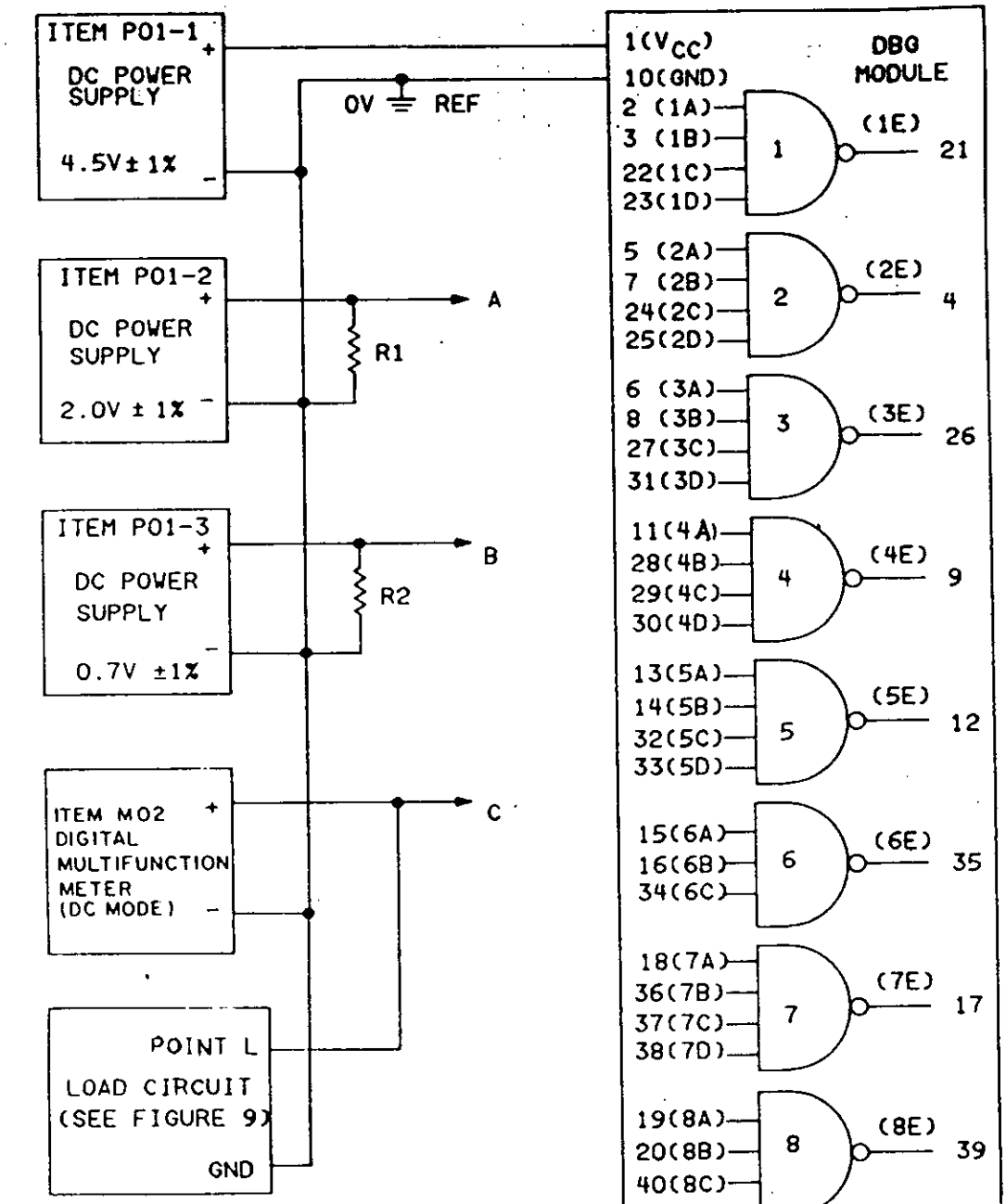
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1E Output voltage, high level	Open	2	21	2.4	4.5	V	8
E2	1E		3	21				
E3	1E		22	21				
E4	1E		23	21				
E5	2E		5	4				
E6	2E		7	4				
E7	2E		24	4				
E8	2E		25	4				
E9	3E		6	26				
E10	3E		8	26				
E11	3E		27	26				
E12	3E		31	26				
E13	4E		11	9				
E14	4E		28	9				
E15	4E		29	9				
E16	4E		30	9				
E17	5E		13	12				
E18	5E		14	12				
E19	5E		32	12				
E20	5E		33	12				
E21	6E		15	35				
E22	6E		16	35				
E23	6E		34	35				

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TABLE VII. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E24	7E Output voltage, high level	Open	18	17	2.4	4.5	V	8
E25	7E		36	17				
E26	7E		37	17				
E27	7E		38	17				
E28	8E		19	39				
E29	8E		20	39				
E30	8E		40	39				

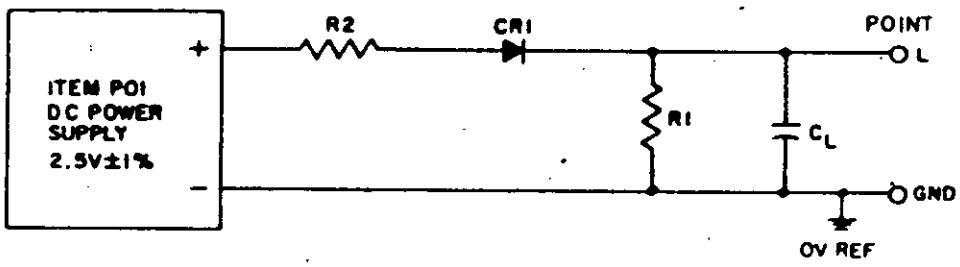
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Since items PO1-2 and PO1-3 are not used simultaneously, only two item PO1 power supplies are required.

FIGURE 3. Output voltage (low-level and high-level) test circuit for table VI and VII.

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NOTES:
1. CR1 = 1N916, or equivalent.
2. R1 = 23.7 kΩ ±1%, 1/8 W for a current flow of 100 µA, into point L when point L is held at 2.4 V.
3. R2 = Select value for a current flow of 2.0 mA ±5% out of point L when point L is held at 0.3 V.
4. C_L = 40 ±5 pF (including probe capacitance) for test conditions K1-K8, L1-L8, M1-M8 and N1-N8.
   - 140 ±5 pF (including probe capacitance) for test conditions K9-K16, L9-L16, M9-M16 and N9-N16.
   - 50 ±5 pF (including probe capacitance) for test conditions P1-P8 and R1-R8.
5. C_L is not required for dc electrical tests D and E (table II).
6. The maximum lead length from point "L" to the appropriate module pin shall be less than 1 inch (25.4 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 9. Logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1A Input current, low level	Open	2	Open	3,22,23	50	200	µA	10
F2	1B		3		2,22,23				
F3	1C		22		2,3,23				
F4	1D		23		2,3,22				
F5	2A		5		7,24,25				
F6	2B		7		5,24,25				
F7	2C		24		5,7,25				
F8	2D		25		5,7,24				
F9	3A		6		8,27,31				
F10	3B		8		6,27,31				
F11	3C		27		6,8,31				
F12	3D		31		6,8,27				
F13	4A		11		28,29,30				
F14	4B		28		11,29,30				
F15	4C		29		11,28,30				
F16	4D		30		11,28,29				
F17	5A		13		14,32,33				
F18	5B		14		13,32,33				
F19	5C		32		13,14,33				
F20	5D		33		13,14,32				
F21	6A		15		16,34				
F22	6B		16		15,34				
F23	6C		34		15,16				

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TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F24	7A Input current, low level	Open	18	Open	36,37,38	50	200	μA	10
F25	7B	↓	36	↓	18,37,38	↓	↓	↓	↓
F26	7C		37		18,36,38				
F27	7D		38		18,36,37				
F28	8A		19		20,40				
F29	8B		20		19,40				
F30	8C		40		19,20				

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1A Input current, high level	2	Open	3,22,23	Open	0	10	μA	10
G2	1B	3		2,22,23					
G3	1C	22		2,3,23					
G4	1D	23		2,3,22					
G5	2A	5		7,24,25					
G6	2B	7		5,24,25					
G7	2C	24		5,7,25					
G8	2D	25		5,7,24					
G9	3A	6		8,27,31					
G10	3B	8		6,27,31					
G11	3C	27		6,8,31					
G12	3D	31		6,8,27					
G13	4A	11		28,29,30					
G14	4B	28		11,29,30					
G15	4C	29		11,28,30					
G16	4D	30		11,28,29					
G17	5A	13		14,32,33					
G18	5B	14		13,32,33					
G19	5C	32		13,14,33					
G20	5D	33		13,14,32					
G21	6A	15		16,34					
G22	6B	16		15,34					
G23	6C	34		15,16					
G24	7A	18		36,37,38					

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TABLE IX. Input current (high-level) test - Continued.

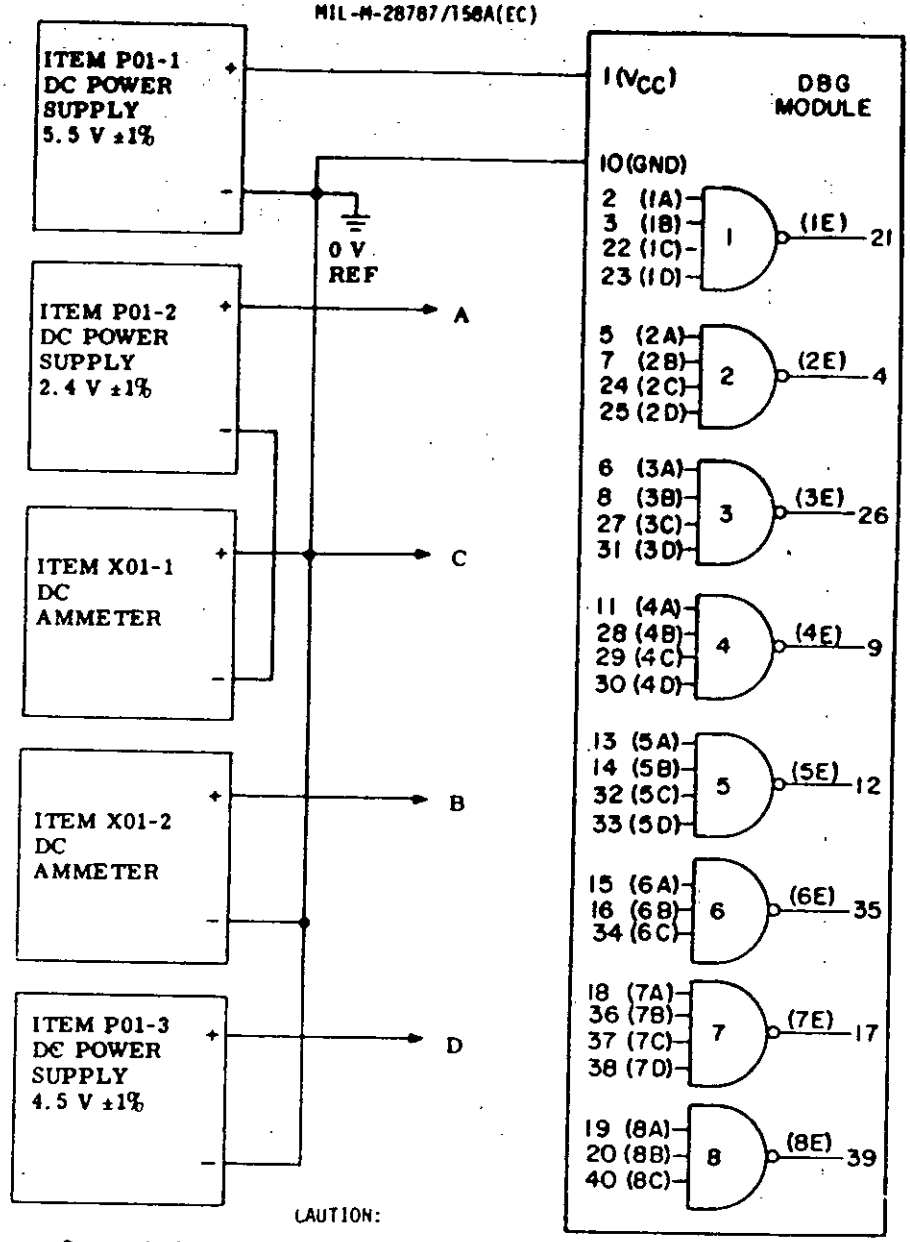
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G25	7B Input current, high level	36	Open	18,37,38	Open	0	10	μA	10
G26	7C	37		18,36,38					
G27	7D	38		18,36,37					
G28	8A	19		20,40					
G29	8B	20		19,40					
G30	8C	40		19,20					

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1E Output short circuit current	Open	21	2,3,22,23	Open	3	15	mA	10
H2	2E		4	5,7,24,25					
H3	3E		26	6,8,27,31					
H4	4E		9	11,28,29,30					
H5	5E		12	13,14,32,33					
H6	6E		35	15,16,34					
H7	7E		17	18,36,37,38					
H8	8E		39	19,20,40					

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:

1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01, is required.
2. Since items P01-2 and P01-3 are not used simultaneously, only two item P01 power supplies are required.
3. With point "A" disconnected, item X01-1 should indicate zero current on the 1 µA range.
4. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for table VIII, IX and X.

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TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Power supply current, low level	C	Open	A	0.5	4.1	mA	11
J2	Power supply current, high level	B	A	Open	0.2	1.6	mA	11

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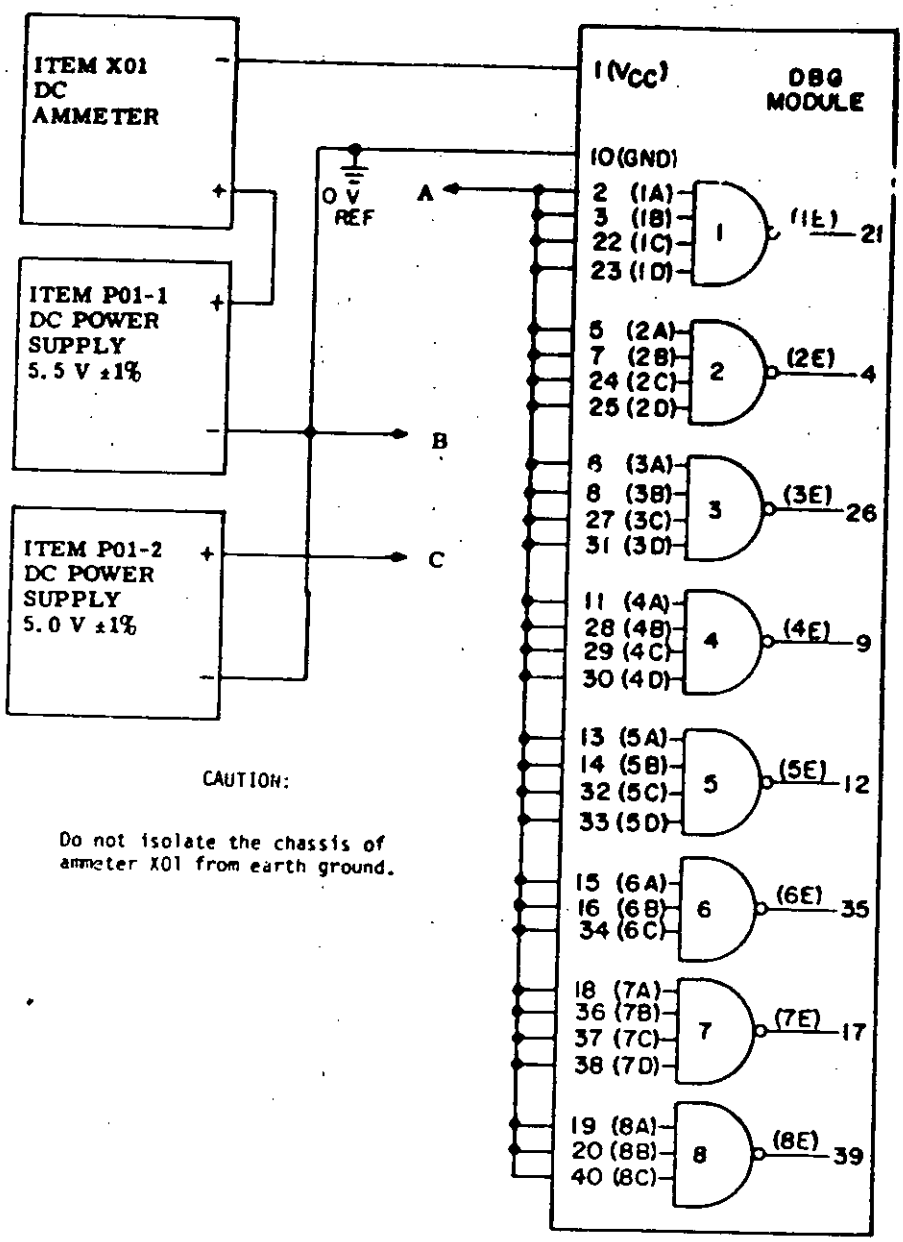


FIGURE 11. Power supply current (low-level and high-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1E Transition time (t_{TLH}), C_L = 40 pF	2	3,22,23	21	5	60	ns	12, 13
K2	2E	5	7,24,25	4				
K3	3E	6	8,27,31	26				
K4	4E	11	28,29,30	9				
K5	5E	13	14,32,33	12				
K6	6E	15	16,34	35				
K7	7E	18	36,37,38	17				
K8	8E	19	20,40	39				
K9	1E Transition time (t_{TLH}), C_L = 140 pF	2	3,22,23	21	10	100		
K10	2E	5	7,24,25	4				
K11	3E	6	8,27,31	26				
K12	4E	11	28,29,30	9				
K13	5E	13	14,32,33	12				
K14	6E	15	16,34	35				
K15	7E	18	36,37,38	17				
K16	8E	19	20,40	39				

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1E Transition time (t_{THL}), C_L = 40 pF	2	3,22,23	21	2	30	ns	12, 13
L2	2E	5	7,24,25	4				
L3	3E	6	8,27,31	26				
L4	4E	11	28,29,30	9				
L5	5E	13	14,32,33	12				
L6	6E	15	16,34	35				
L7	7E	18	36,37,38	17				
L8	8E	19	20,40	39				
L9	1E Transition time (t_{THL}), C_L = 140 pF	2	3,22,23	21	5	60		
L10	2E	5	7,24,25	4				
L11	3E	6	8,27,31	26				
L12	4E	11	28,29,30	9				
L13	5E	13	14,32,33	12				
L14	6E	15	16,34	35				
L15	7E	18	36,37,38	17				
L16	8E	19	20,40	39				

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TABLE XIV. Propagation delay time (tPLH) test.

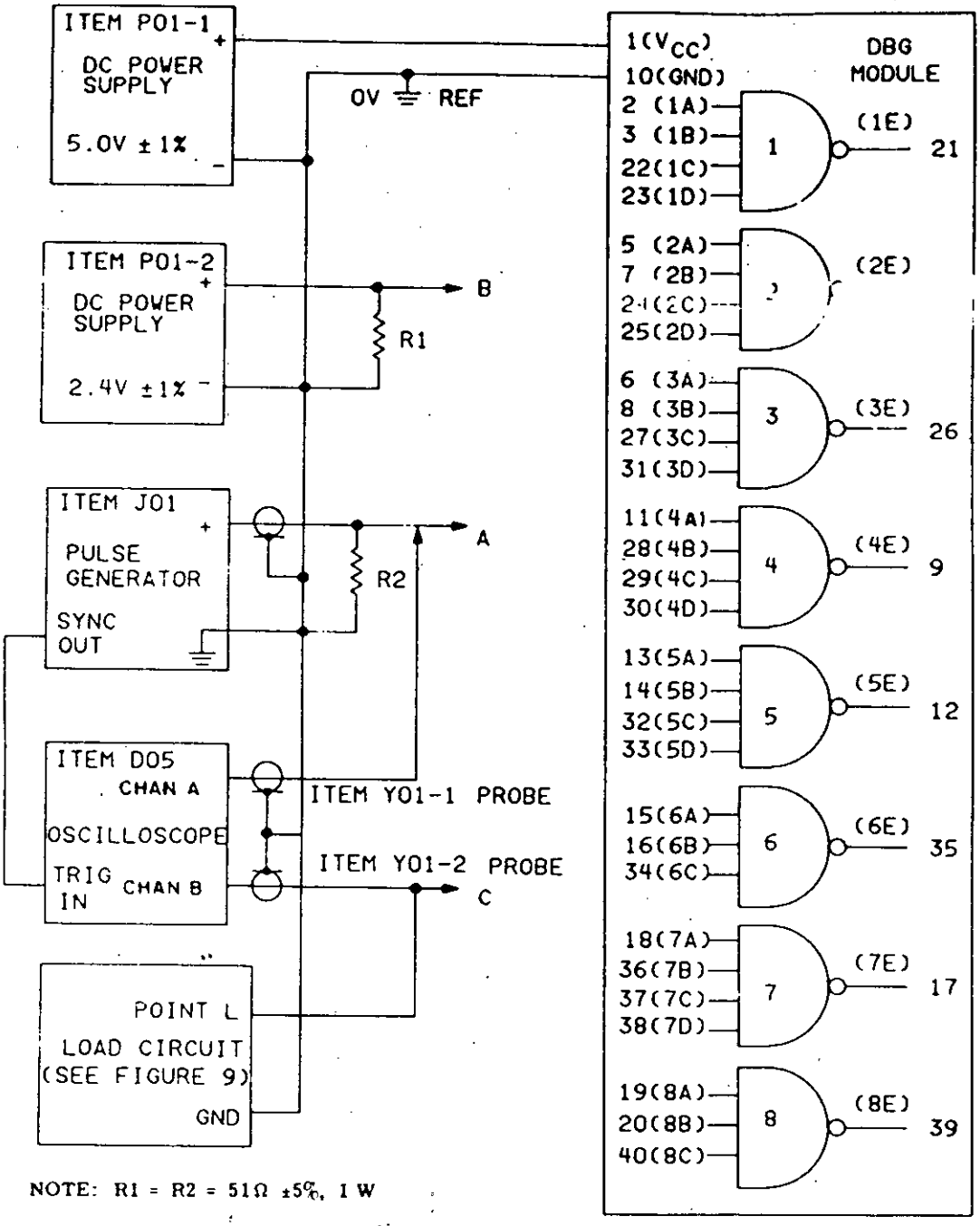
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1E Propagation delay time (tPLH), CL = 40 pF	2	3,22,23	21	10	60	ns	12,13
M2	2E	5	7,24,25	4				
M3	3E	6	8,27,31	26				
M4	4E	11	28,29,30	9				
M5	5E	13	14,32,33	12				
M6	6E	15	16,34	35				
M7	7E	18	36,37,38	17				
M8	8E	19	20,40	39				
M9	1E Propagation delay time (tPLH), CL = 140 pF	2	3,22,23	21	10	90		
M10	2E	5	7,24,25	4				
M11	3E	6	8,27,31	26				
M12	4E	11	28,29,30	9				
M13	5E	13	14,32,33	12				
M14	6E	15	16,34	35				
M15	7E	18	36,37,38	17				
M16	8E	19	20,40	39				

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1E Propagation delay time (t_{PHL}), C_L = 40 pF	2	3,22,23	21	10	66	ns	12,13
N2	2E	5	7,24,25	4				
N3	3E	6	8,27,31	26				
N4	4E	11	28,29,30	9				
N5	5E	13	14,32,33	12				
N6	6E	15	16,34	35				
N7	7E	18	36,37,38	17				
N8	8E	19	20,40	39				
N9	1E Propagation delay time (t_{PHL}), C_L = 140 pF	2	3,22,23	21	10	96		
N10	2E	5	7,24,25	4				
N11	3E	6	8,27,31	26				
N12	4E	11	28,29,30	9				
N13	5E	13	14,32,33	12				
N14	6E	15	16,34	35				
N15	7E	18	36,37,38	17				
N16	8E	19	20,40	39				

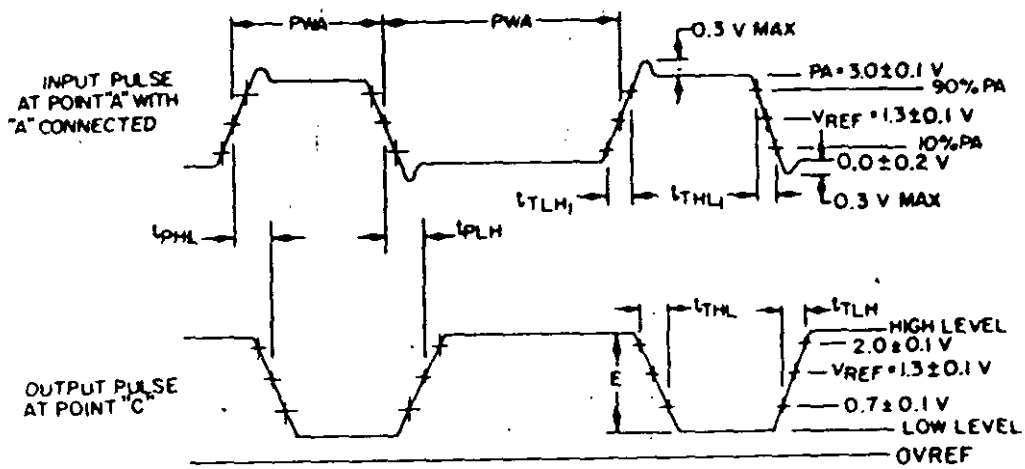
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NOTE: R1 = R2 = 51Ω ±5%, 1 W

FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII through XV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
2. Before making the measurements specified in tables XII, XIII, XIV, and XV, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.3 V.
3. All voltages are referenced to ground.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII through XV.

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TABLE XVI. Output crosstalk (low-level) test.

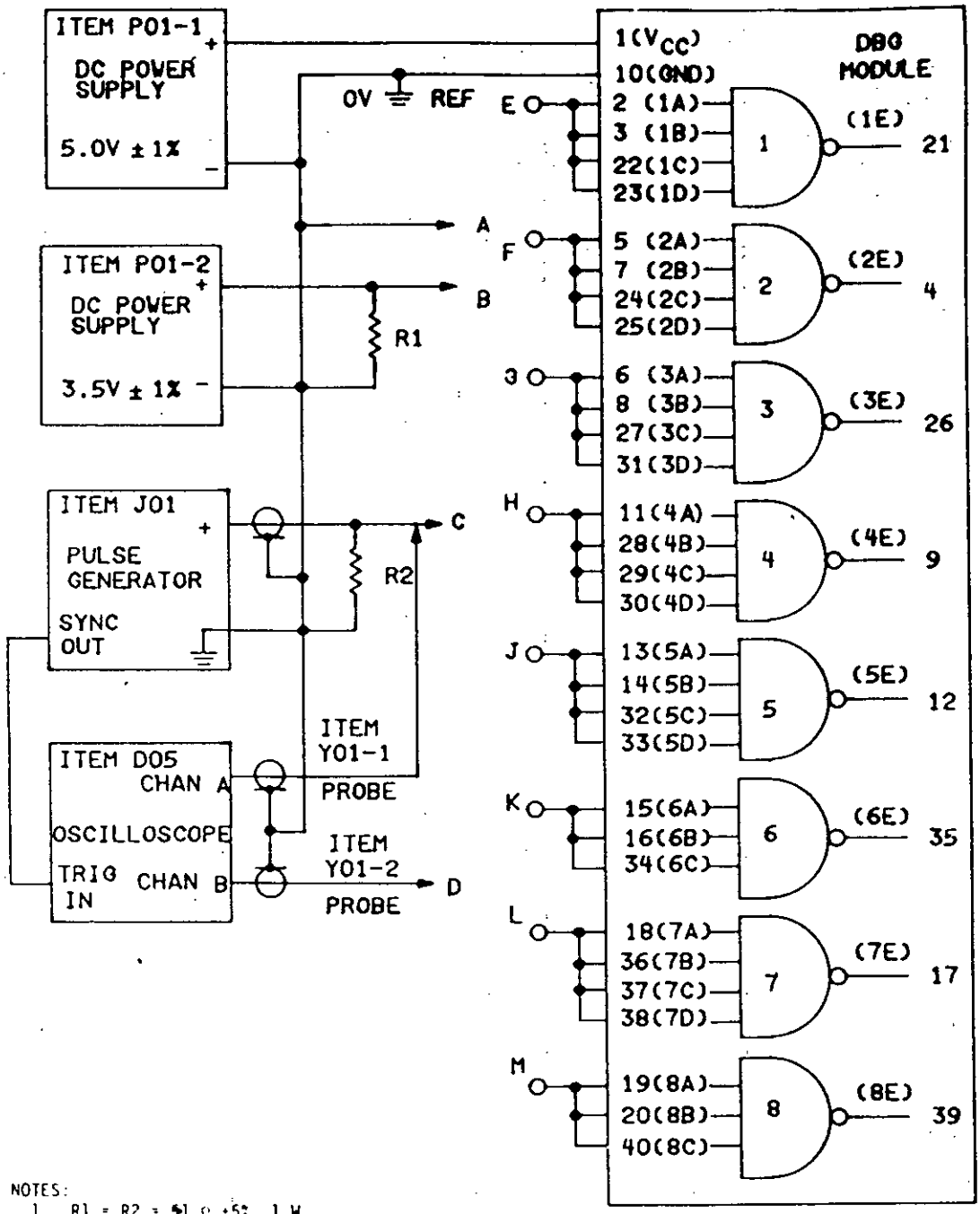
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1E Output crosstalk, low level	Open	E	F,G,H,J,K,L,M	21	0	400	mV	14, 15
P2	2E		F	E,G,H,J,K,L,M	4				
P3	3E		G	E,F,H,J,K,L,M	26				
P4	4E		H	E,F,G,J,K,L,M	9				
P5	5E		J	E,F,G,H,K,L,M	12				
P6	6E		K	E,F,G,H,J,L,M	35				
P7	7E		L	E,F,G,H,J,K,M	17				
P8	8E		M	E,F,G,H,J,K,L	39				

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TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1E Output crosstalk, high level	E	Open	F,G,H,J,K,L,M	21	0	400	mV	14, 15
R2	2E	F		E,G,H,J,K,L,M	4				
R3	3E	G		E,F,H,J,K,L,M	26				
R4	4E	H		E,F,G,J,K,L,M	9				
R5	5E	J		E,F,G,H,K,L,M	12				
R6	6E	K		E,F,G,H,J,L,M	35				
R7	7E	L		E,F,G,H,J,K,M	17				
R8	8E	M		E,F,G,H,J,K,L	39				

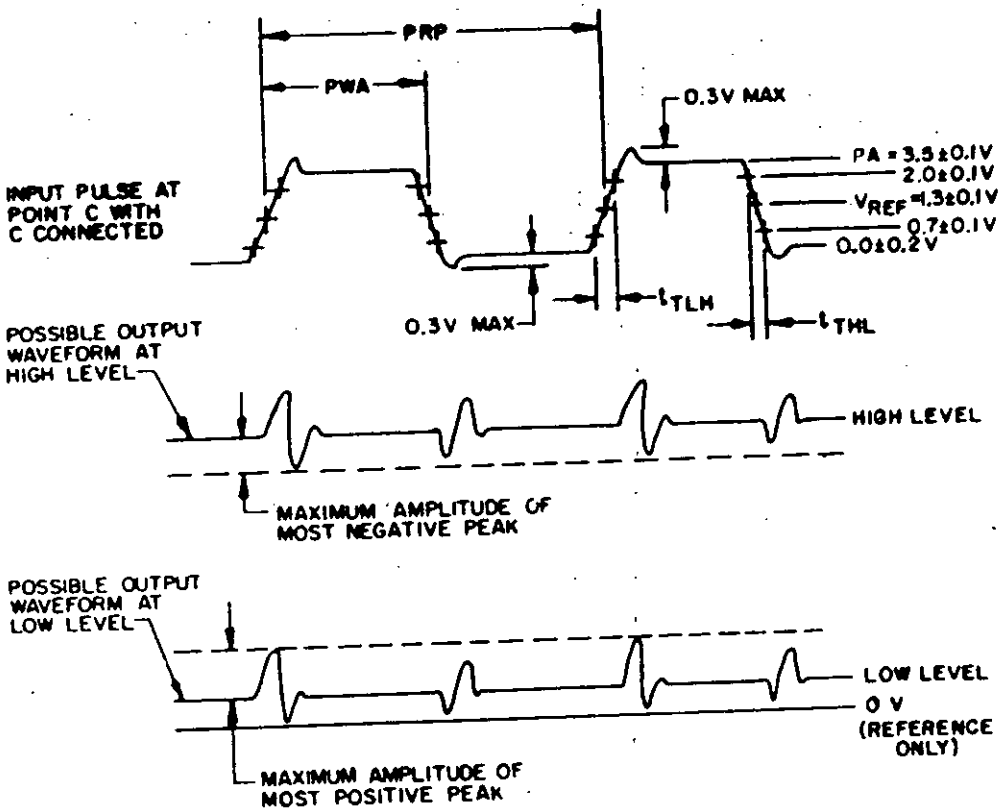
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NOTES:
1. R1 = R2 = 51 Ω ±5%, 1 W.
2. Load all outputs with the load specified on figure 9.

FIGURE 14. Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

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NOTE: Input pulse parameters:
$t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at all inputs).
PWA = 200 ±20 ns.
PRP = 1,000 ±100 ns.

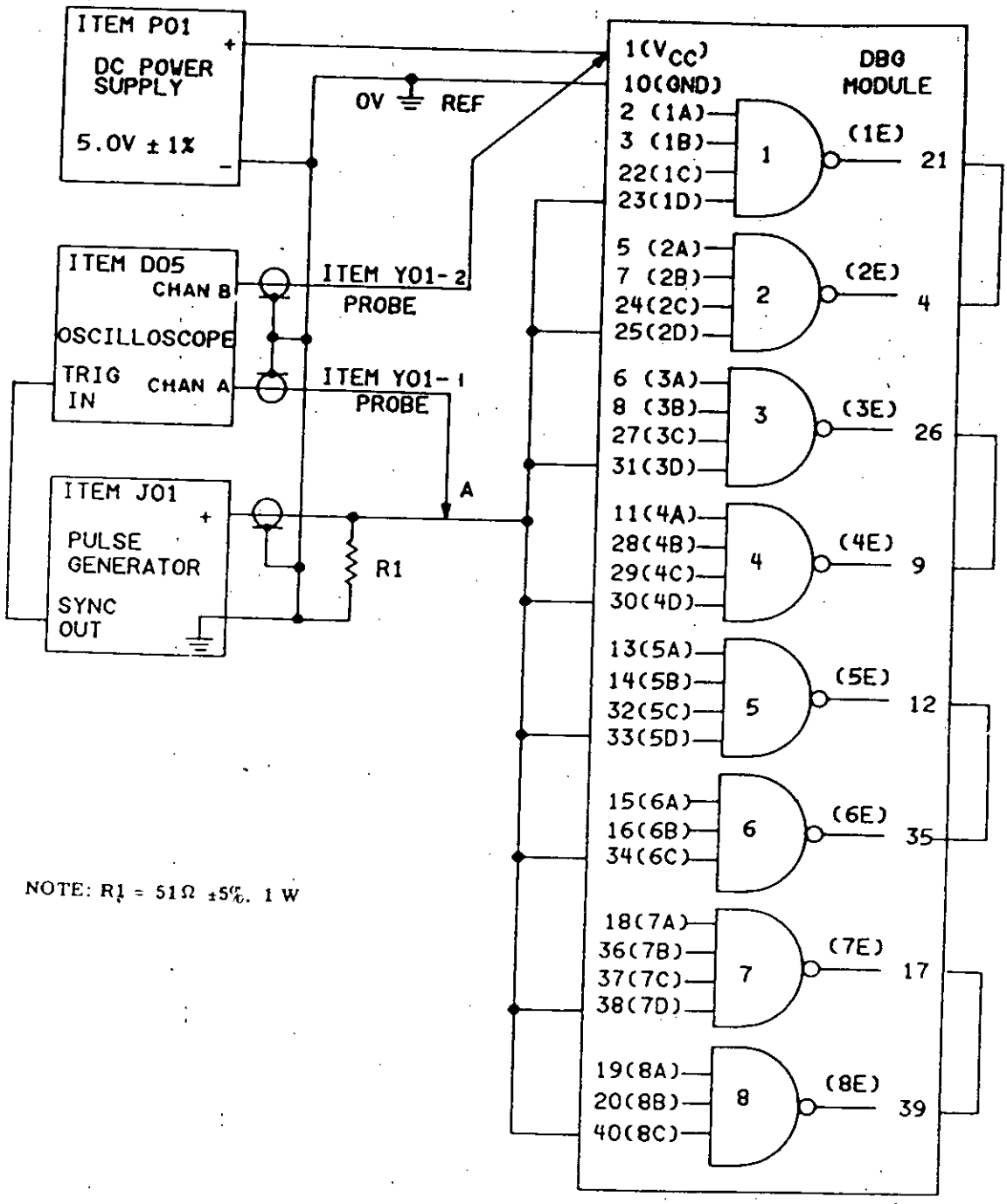
FIGURE 15. Output crosstalk (low-level and high-level) test circuit waveforms for table XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	300	mV	16, 17

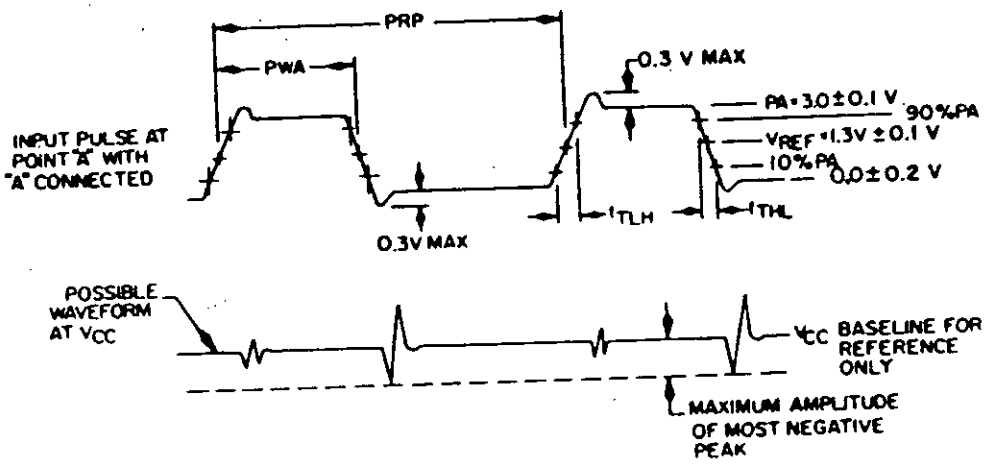
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NOTE: R1 = 51Ω ±5%, 1 W

FIGURE 16. Power supply transient amplitude test circuit for table XVIII.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 20 \pm 2$ ns.
PWA = 200 ±20 ns.
PRP = 1,000 ns ±100 ns.

FIGURE 17. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Power supply filter capacitance	Open	-1	1.65	2.75	µF	18

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TABLE XX. Input capacitance test.

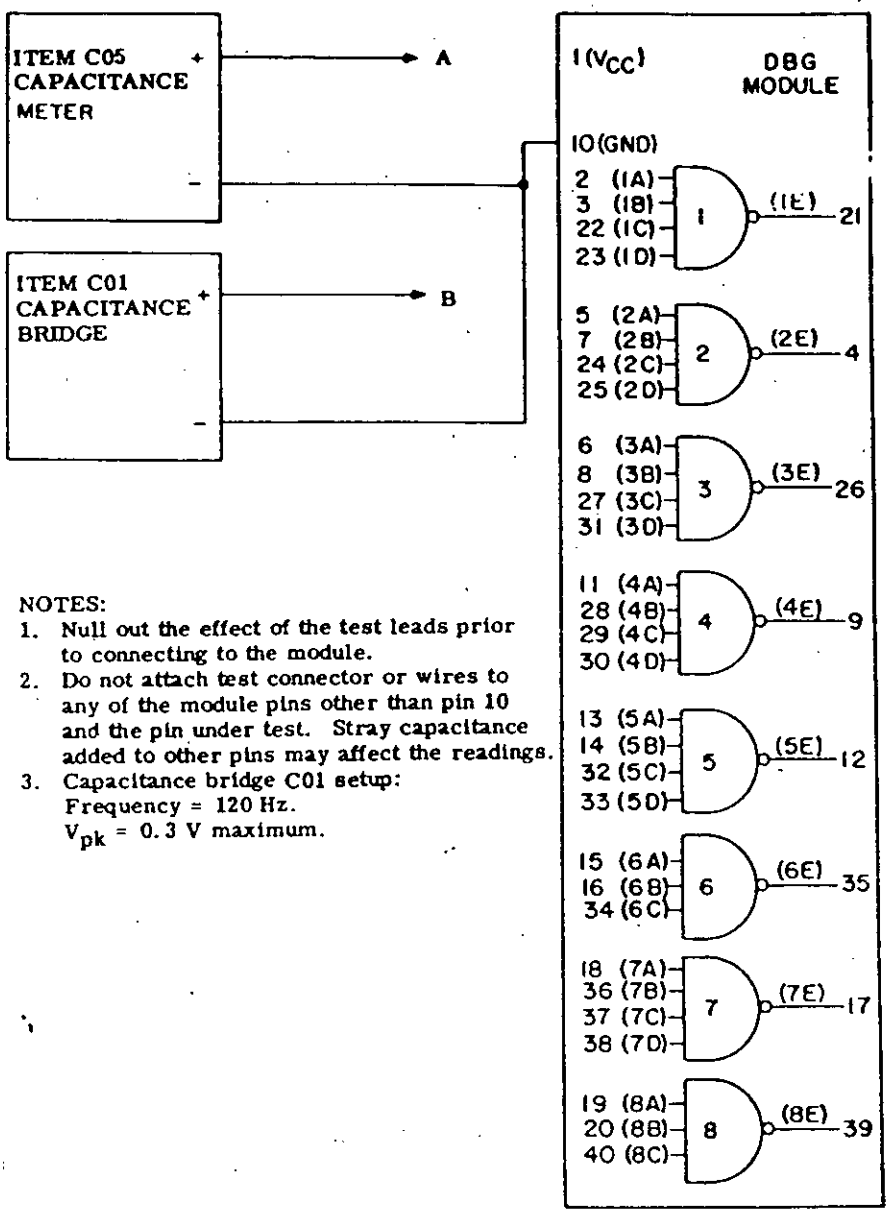
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
U1	1B Input capacitance	3	2	15	pF	18
U2	1A	2				
U3	1C	22				
U4	1D	23				
U5	2A	5				
U6	2B	7				
U7	2C	24				
U8	2D	25				
U9	3A	6				
U10	3C	27				
U11	3B	8				
U12	3D	31				
U13	4A	11				
U14	4B	28				
U15	4C	29				
U16	4D	30				
U17	5A	13				
U18	5B	14				
U19	5C	32				
U20	5D	33				
U21	6A	15				
U22	6B	16				
U23	6C	34				
U24	7A	18				

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
U25	7B Input capacitance	36	2	15	pF	18
U26	7C	37				
U27	7D	38				
U28	8B	20				
U29	8A	19				
U30	8C	40				

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 18. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

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